# Experiment 5

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1. **Aim**: To design and analyze a scalable real-time messenger application (similar to WhatsApp / Facebook Messenger) that allows users to register, send one-to-one and group messages, share media, and receive messages instantly while ensuring high availability, reliability, and low latency using distributed system concepts.

## 2. Objective:

- To identify functional and non-functional requirements of a messaging system
- To design a high-level architecture (HLD) for a large-scale chat application
- To understand real-time communication using WebSockets
- To apply CAP theorem trade-offs in messaging systems
- To understand message queues, storage, caching, and delivery guarantees
- To study how availability and reliability are prioritized in chat systems

## 3. Tools:

- Draw.io – System Architecture & HLD Diagram
- MySQL / PostgreSQL – User & group metadata
- NoSQL DB (MongoDB / Cassandra) – Message storage
- Redis – Online presence & caching
- WebSocket Server – Real-time messaging
- Apache Kafka / Message Queue – Asynchronous delivery
- Object Storage (S3 equivalent) – Media files
- Load Balancer – Traffic distribution

## 4. System Requirements:

### A. Functional Requirements:-

1) Client should be able to register and login securely

2) System should support one-to-one messaging

3) System should support group messaging
4) System should support text and media messages (image/video/file)
5) System should preserve chat history
6) Client should see message delivery and read receipts

7) Client should view chat list and conversation history
8) System should support creating and managing groups

## B. Non-Functional Requirements:-

**1. Scalability**

- Target Users: 1 Billion users
- Avg Messages/User/Day: 100
- Total Messages: 100 Billion/day
- Storage Estimate: ~100 TB messages

System must handle millions of concurrent connections.

**2. Availability vs Consistency (CAP Theorem)** Availability

>>> Consistency (Eventual Consistency)

**Reason:**

- Messages must be delivered even during failures
- Slight delay in read receipts is acceptable
- Losing messages is unacceptable

**Example:**

- Message delayed → Acceptable
- Message lost → Critical failure **3. Latency**

Target latency: 200–300 ms
Messages should appear almost instantly.

**4. Reliability**

- No message loss
- Guaranteed delivery

- Retry & acknowledgment mechanism

## 5. High Level Design (HLD):

The system follows a microservices-based architecture:

**API Gateway**

- Authentication, routing, rate limiting
- Entry point for all clients

**User Service**

- Registration & login
- Profile & contacts
- Stored in SQL DB

**Messaging Service**

- Handles sending/receiving messages
- Stores messages in NoSQL DB
- Maintains conversation threads

**Real-Time WebSocket Server**

- Persistent connections with clients
- Instant message delivery
- Presence & typing indicators

**Group Service**

- Group creation & membership
- Group metadata storage

**Media Service**

- Upload & download media files
- Stores media in object storage

**Message Queue (Kafka)**

- Asynchronous message delivery

- Decouples sender & receiver
- Ensures reliability

**Redis Cache**

- Online/offline status
- Recent chats
- Unread counts

**Notification Service** • Push

notifications for offline users

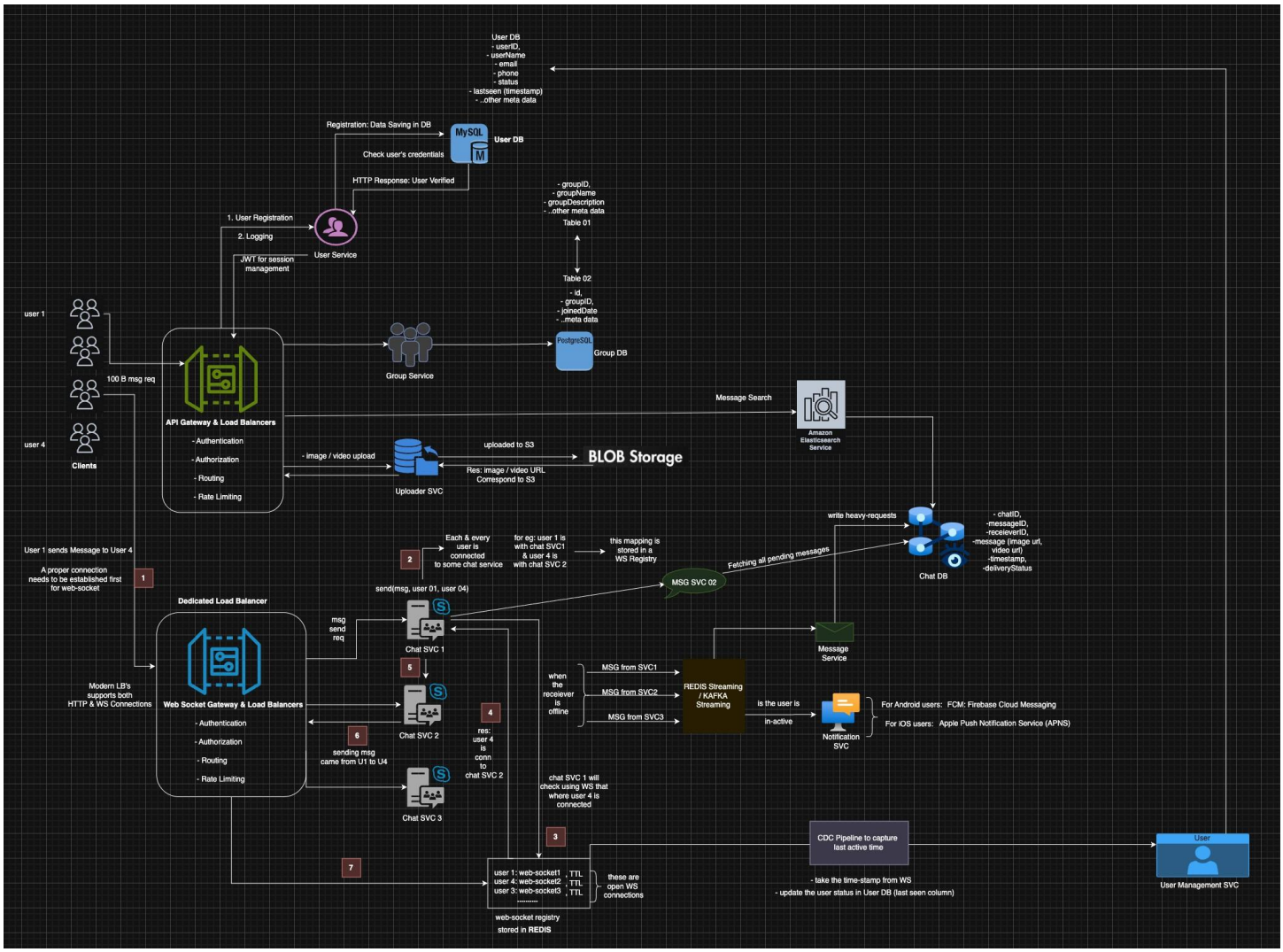
## 6. Messaging Workflow:

Step-1: User sends message
Step-2: Client → WebSocket Server

Step-3: Message stored in DB
Step-4: Message pushed to receiver if online
Step-5: If offline → Queue + Notification Step-6:
Receiver reads → Read receipt sent

## 7. Scalability Solution:

- Horizontal scaling of WebSocket servers
- Sharding messages by user ID
- Kafka queues for async delivery
- Redis caching for presence
- NoSQL DB for massive writes
- Load balancer distributes connections
- Stateless microservices


User DB
- userID,
- userName
- email
- phone
- status
- lastseen (timestamp)
- ..other meta data
Registration: Data Saving in DB
MySQL
User DB
Check user's credentials
HTTP Response: User Verified
1. User Registration
2. Logging
JWT for session management
User Service
- groupID,
- groupName
- groupDescription
- ..other meta data
Table 01
Table 02
- id,
- groupID,
- joinedDate
- ..meta data
user 1
100 B msg req
Group Service
PostgreSQL
Group DB
API Gateway & Load Balancers
- Authentication
- Authorization
- Routing
- Rate Limiting
Clients
user 4
Message Search
Amazon Elasticsearch Service
- image / video upload
uploaded to S3
BLOB Storage
Res: image / video URL Correspond to S3
Uploader SVC
write heavy-requests
- chatID,
-messageID,
-receieverID,
-message (image url, video url)
-timestamp,
-deliveryStatus
Chat DB
User 1 sends Message to User 4
A proper connection needs to be established first for web-socket
1
2
Each & every user is connected to some chat service
for eg: user 1 is with chat SVC1 & user 4 is with chat SVC 2
this mapping is stored in a WS Registry
Fetching all pending messages
MSG SVC 02
send(msg, user 01, user 04)
Dedicated Load Balancer
msg send req
Chat SVC 1
Message Service
5
Modern LB's supports both HTTP & WS Connections
Web Socket Gateway & Load Balancers
- Authentication
- Authorization
- Routing
- Rate Limiting
Chat SVC 2
4
6
sending msg came from U1 to U4
res: user 4 is conn to chat SVC 2
Chat SVC 3
when the receiever is offline
MSG from SVC1
MSG from SVC2
MSG from SVC3
REDIS Streaming / KAFKA Streaming
is the user is
in-active
Notification SVC
For Android users: FCM: Firebase Cloud Messaging
For iOS users: Apple Push Notification Service (APNS)
chat SVC 1 will check using WS that where user 4 is connected
3
7
user 1: web-socket1 , TTL
user 4: web-socket2 , TTL
user 3: web-socket3 , TTL
.....
these are open WS connections
web-socket registry stored in REDIS
CDC Pipeline to capture last active time
- take the time-stamp from WS
- update the user status in User DB (last seen column)
User
User Management SVC

## 8. Learning Outcomes (What I Have Learnt):

- Understood how real-time messaging apps like WhatsApp are designed
- Learned difference between functional & non-functional requirements
- Gained understanding of WebSockets & real-time communication
- Learned CAP theorem trade-offs in chat systems
- Understood message queues & delivery guarantees
- Learned how to design highly available messaging systems